Chapter 1

Introduction

1.1 Introduction

1.1.1 Definitions

Lets list some definitions of Tensor. A Tensor T can be regarded as an element of a tensor product space

$$T \in \bigotimes_{j \in J} V^j$$

and we say T is a $|J|$-legged tensor.

Given finite dimensional Vector spaces V and W, we have the identity that $V \otimes W^* \equiv Hom(W, V)$. Now, if we have a Tensor $T \in V_1 \otimes \cdots \otimes V_n$, and since for any Vector space V, $(V^*)^* = V$, we can peel off any number of V_is, take their duals and represent T as a map

$$T : V_n^* \otimes \cdots \otimes V_{k+1}^* \to V_1 \otimes \cdots \otimes V_k$$

Note that we present the domain here in a contravariant fashion due to the contravariance of the dual.

Lets look at an example, let $t : V_1 \otimes V_2 \otimes W_2^* \otimes W_1^*$. We can represent as an operator in dirac notation:

$$t = \sum_{i_1, i_2, j_1, j_2} T_{j1j2i1i2} \left| v_{i_1} \right\rangle \left| v_{i_2} \right\rangle \left\langle w_{j_1} \right| \left\langle w_{j_2} \right|$$

Graphically, we represent a general tensor

$$T : (W_n^* \otimes \cdots \otimes W_1) \to (V_1 \otimes V_2 \otimes \cdots \otimes V_n)$$

as follows, where we choose incoming arrows for duals in the domain and outgoings for normal spaces, and on the bottom we choose incoming for duals and outgoing for normals.

1.1.2 Review of Symmetric Tensor networks.

Generally, we follow the notations of [SV12], [Sch+20], [Sch+20], [Sch20] which gave an overview for implementing $SU(2)$ invariant tensor networks. A general overview for (Non)abelian symmetries is [SPV10], and a detailed account for the abelian $U(1)$ in [SPV11]. There are a multitude of implementations given in major packages, such as ITensor, [FWS22], Cyten, TenPy, [Hau+24], QSpace, [Wei24] ChemTensor, UniTensor, YasTn (Abelian Only) [Ram+25] (MPSKit?), (Cytnx (Abelian only i think.))). A general account of symmetries is given in [Unf24] with its respective implementation in [Hau+24] (We have to build cytens citation manually.)

1.1.3 Quantum Gates as Tensor Network Operators

As a sanity check, view this in context of a quantum circuit. A quantum circuit is defined as (unitary) map $(\mathbb{C}^2)^n \to (\mathbb{C}^2)^m$.

$$CNOT : C^2 \otimes C^2 \to C^2 \otimes C^2 = \sum_{ij} |i, i \oplus j\rangle \langle i, j|$$

$$H = \frac{1}{\sqrt{2}} \sum_{ij} (-1)^{ij} |i\rangle \langle j|$$

1.2 Symmetries

What do we know now? First of all, we take a symmetry group G that is finite and compact (why?)[For compact groups, representations are completely reducible (Maschke) and the irreps are finite aswell. Finite groups are contained in compact groups.]. Then we look for its unitary representation $U(G)$ on a Hilbert space H. Furhtermore, if this is a multiparty hilbert space, lets say n sites, we view the unitary representations U_n at site n. The unitary representation $U_n : G \to Hom(V_n, V_n)$ is a covariant linear unitary map. A map $V_i \to V_j$ is symmetric if it is an intertwiner of the representations, i.e. the following diagram commutes:

$$\begin{tikzcd} V_i \arrow[r, "U_{V_i}(g)"] \arrow[d, "f"'] & V_i \arrow[d, "f"] \\ V_j \arrow[r, "U_{V_j}(g)"'] & V_j \end{tikzcd}$$

Note that here V_i is called a *symmetry space*.

Some examples of these groups include $\mathrm{U}(1), \mathrm{SU}(2), \mathbb{Z}_n$

Now where the intutition is more shaky: Consider a symmetry space V_i on a single site (why single). Then we get that it decomposes into a direct sum of irreps V_i^a,

$$V_i = \bigoplus_a \bigoplus_{i=1}^{d_a} V_i^a$$

and we can easily construct an isomporphism (but this depends on choosing a basis for D_a.)

$$V_i = \bigoplus_a \bigoplus_{i=1}^{d_a} V_i^a = \bigoplus_a (D_a \otimes V_i^a).$$

Well, the isom we can construct here would look like $\bigoplus_1^{d_a} V_i^a \to D_a \otimes V_i^a$.

Consider a single site of 3 spin 1/2 particles. Under SU(2) this decomposes as

$$2 \otimes 2 \otimes 2 = 4 \oplus 2 \oplus 2$$

Now consider that C^2 is just the fundamental representation of SU(2).

Note here, an invariant subspace $W \subseteq V$ is a subset of vectors s.t. for all $U \in \mathrm{SU}(2)$, $Uw \in W$.

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